

AE 8883: Spacecraft Engineering I

Embedded Systems

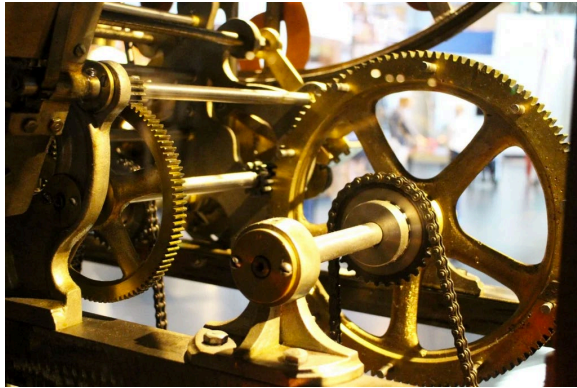
October 3, 2022



Sterling Peet

What is an Embedded System?

What is a System?



A system is a set of interrelated parts or components, working together (to perform a specific task)

What does Embedded mean?

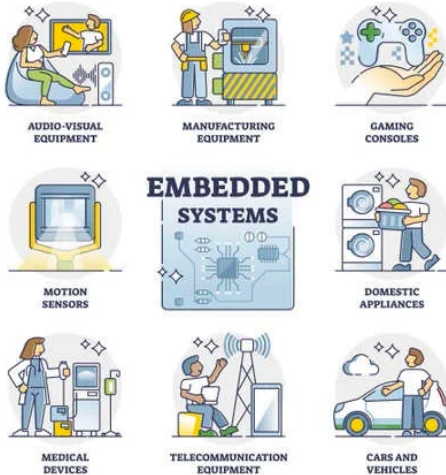


Fixed firmly in a surrounding [system]; implanted

Integrated in a non-trivial way

What is an Embedded System?

Embedded Systems



A combination of hardware and software, and often additional mechanical systems, designed to perform a dedicated function

General Purpose Computer



A combination of hardware and software that serves as a general purpose computing platform

What is an Embedded System?

Sunbeam Deluxe Automatic
Radiant Control Toaster



VS

Oster 6329



What is an Embedded System?

Sunbeam Deluxe Automatic Radiant Control Toaster

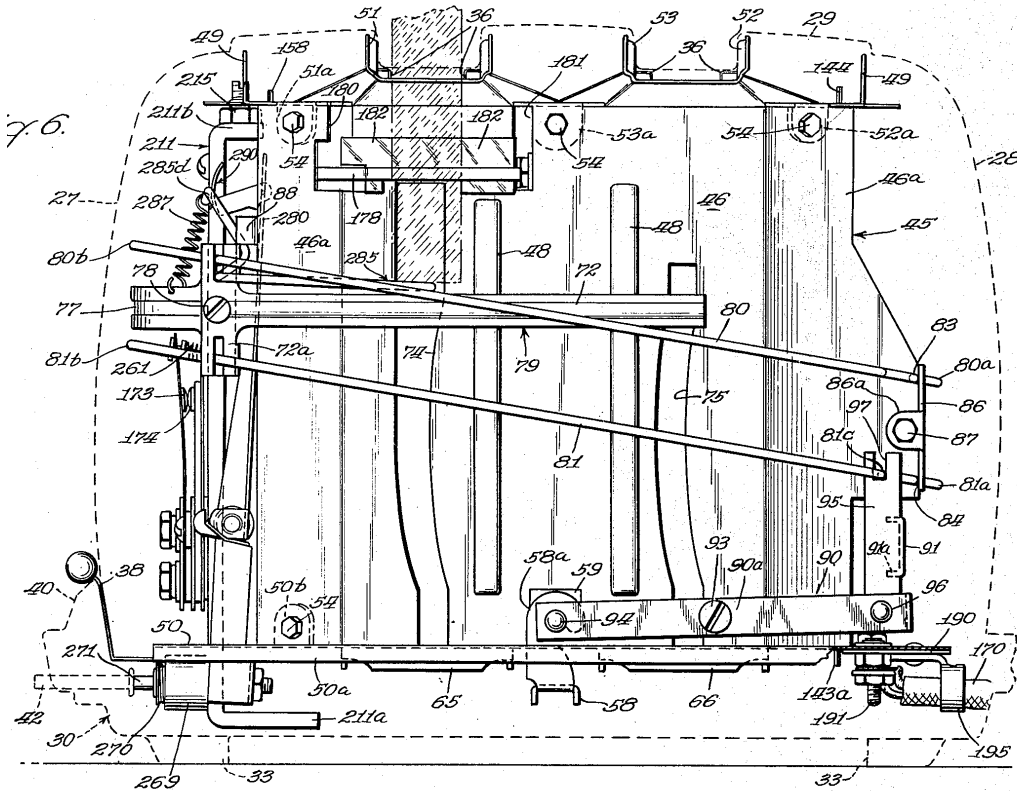
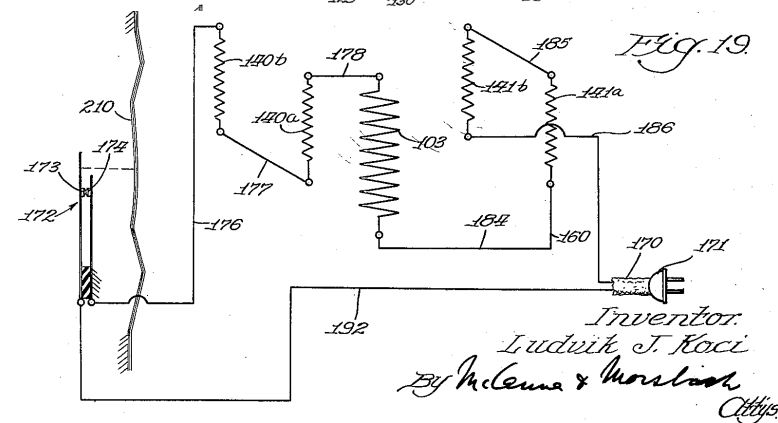
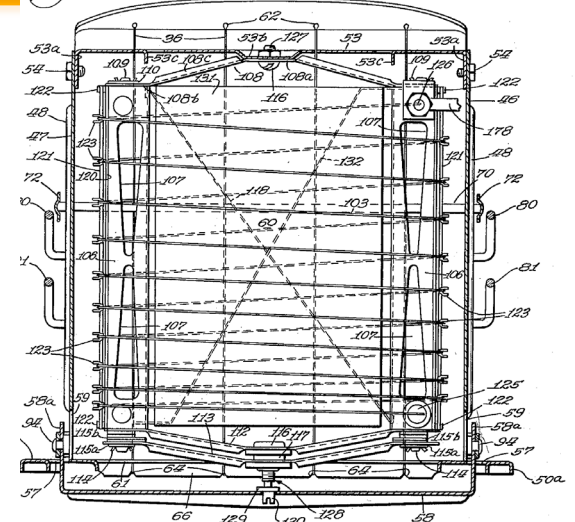


Fig. 8.



What is an Embedded System?

Sunbeam Deluxe Automatic Radiant Control Toaster

NOT AN EMBEDDED SYSTEM

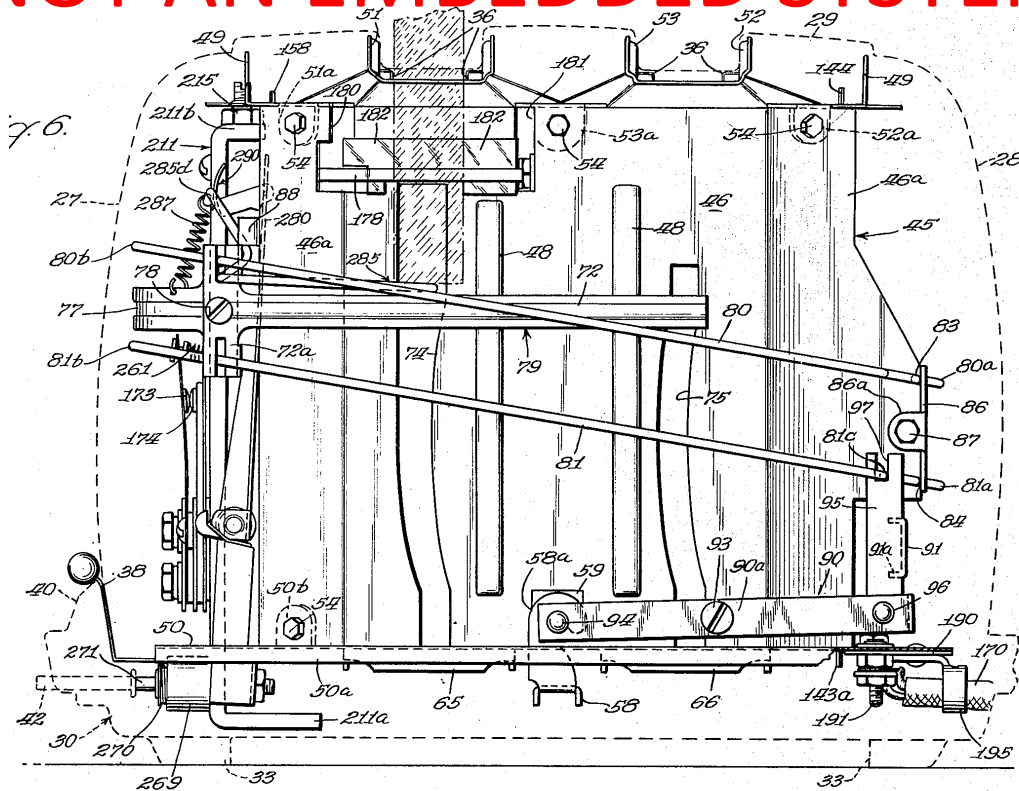
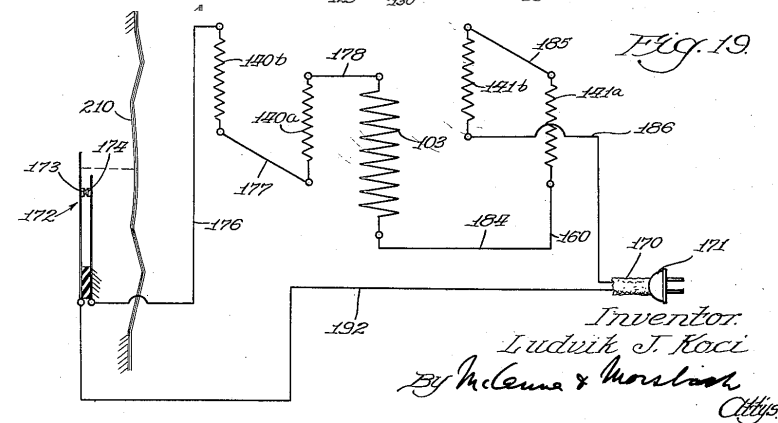
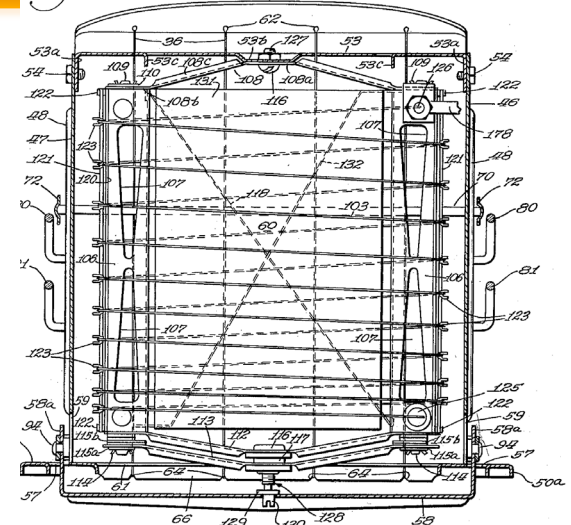


Fig. 8.



Inventor:
Ludvik J. Koci
By McName & Morlock
Attys

What is an Embedded System?

Sunbeam Deluxe Automatic
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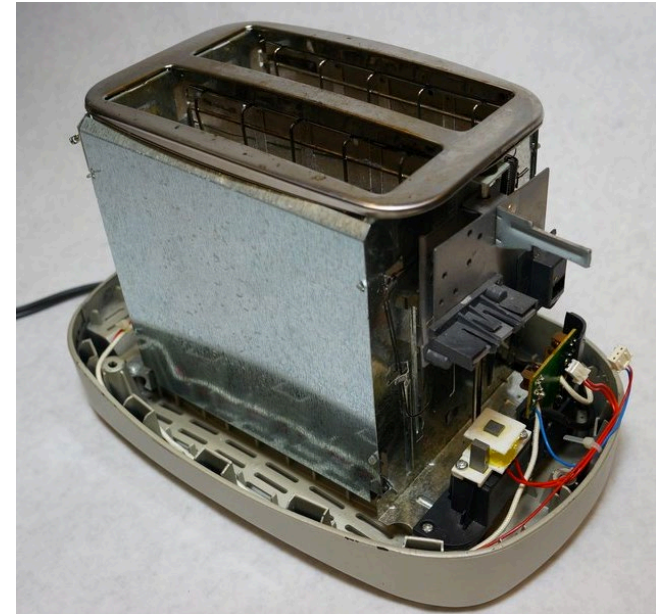
What is an Embedded System?

Oster 6329



What is an Embedded System?

Oster 6329 Insides



What is an Embedded System?

Sunbeam Deluxe Automatic
Radiant Control Toaster



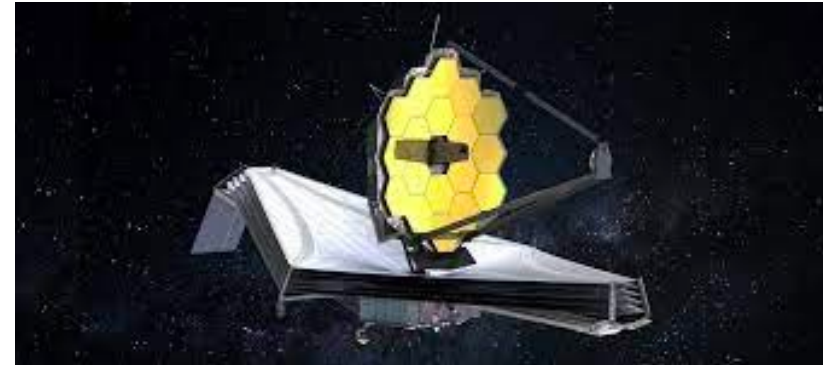
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Where Do We Find Embedded Systems?

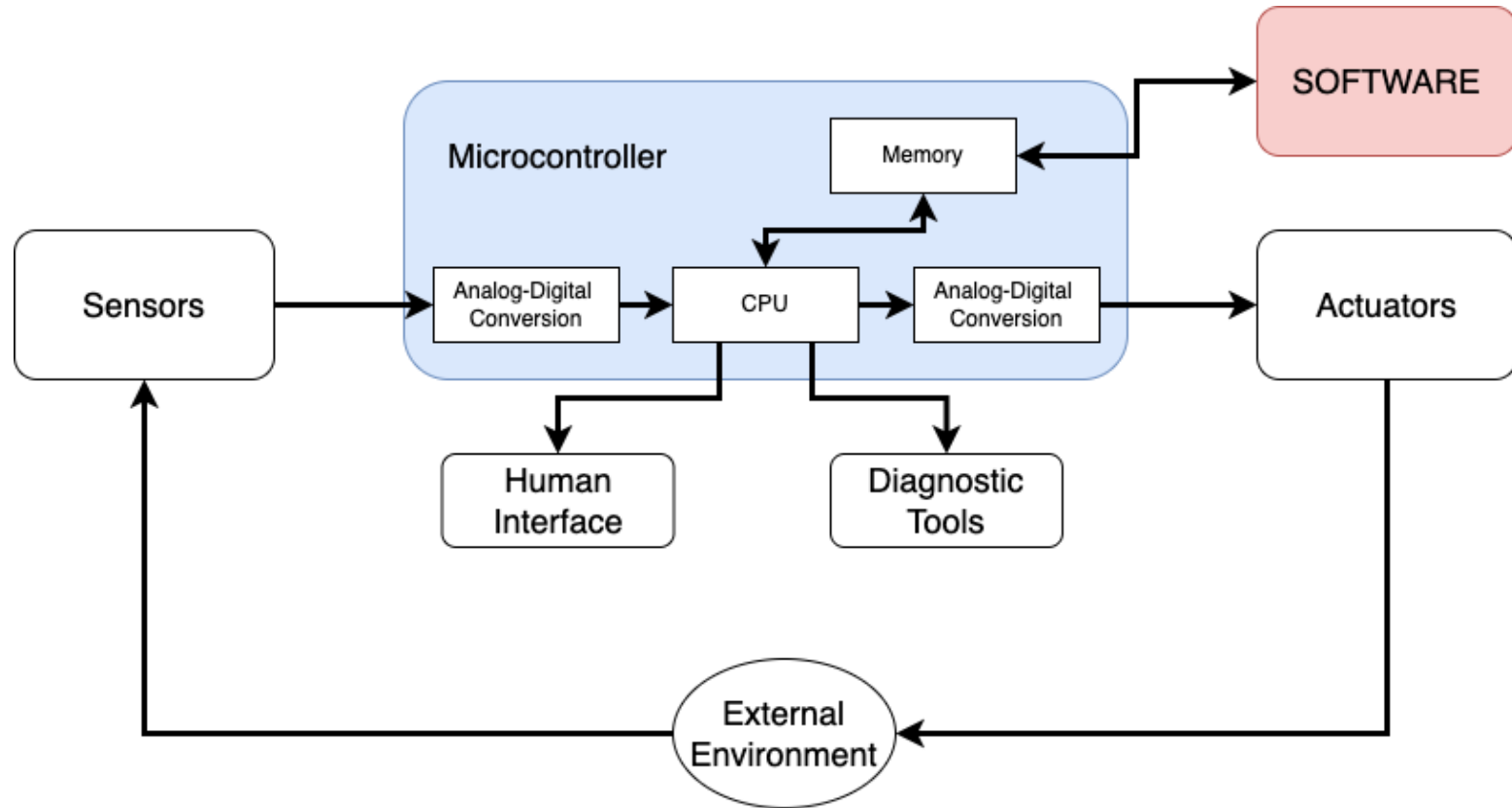
- Digital Watches
- Game Consoles
- Digital Cameras
- Wireless Routers
- Televisions
- Microwave Ovens
- Calculators
- Vending Machines
- **Spacecraft!**
- Home Appliances
- Transportation
- Communication
- Health Care
- Business Applications & Offices
- Defense Applications
- Aerospace Applications
- Smart Agricultural Applications



Typical Characteristics of Good Embedded Systems

- Performs specific tasks or functions
- Time bound operations
- Low power requirements for operation
- Efficiency is a design consideration
- Minimal user interface
- Highly responsive user interface
- Minimal human intervention during operation
- Highly stable operation with little to no maintenance required
- Highly reliable operation
- Uses programmable microcontroller[s] for core functionality

What makes up an Embedded System?



Types of Embedded Platforms

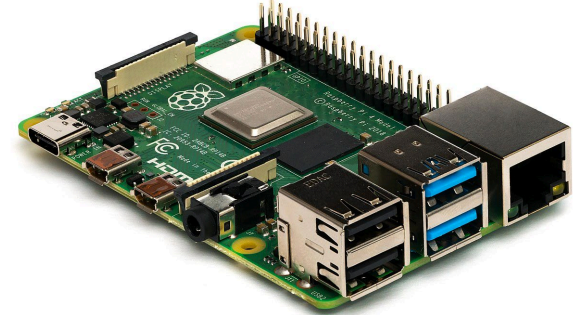
- **Microprocessor / Microcontroller**
 - Most generic, normal computational processor with General Purpose IO pins (GPIO)
- **Media Processor**
 - Hardware supported Video and Audio Codecs
- **Digital Signal Processor (DSP)**
 - Hardware support of higher order math for signal processing (like FFTs)
- **Application Specific Integrated Circuit (ASIC)**
 - Special runs of integrated circuits that are not commonly available or support odd requirements (usually mixed analog and digital)
- **Field Programmable Gate Array (FPGA)**
 - End product engineer can configure the digital logic as desired

Common Platforms for Embedded Systems

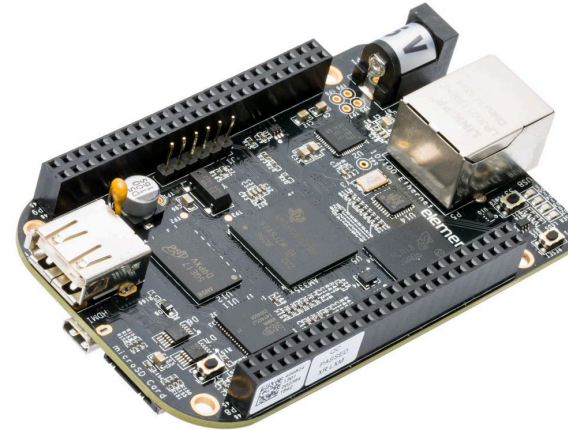
Arduino



Raspberry Pi



BeagleBone



Parts of a Microcontroller

CPU: "The Brain"

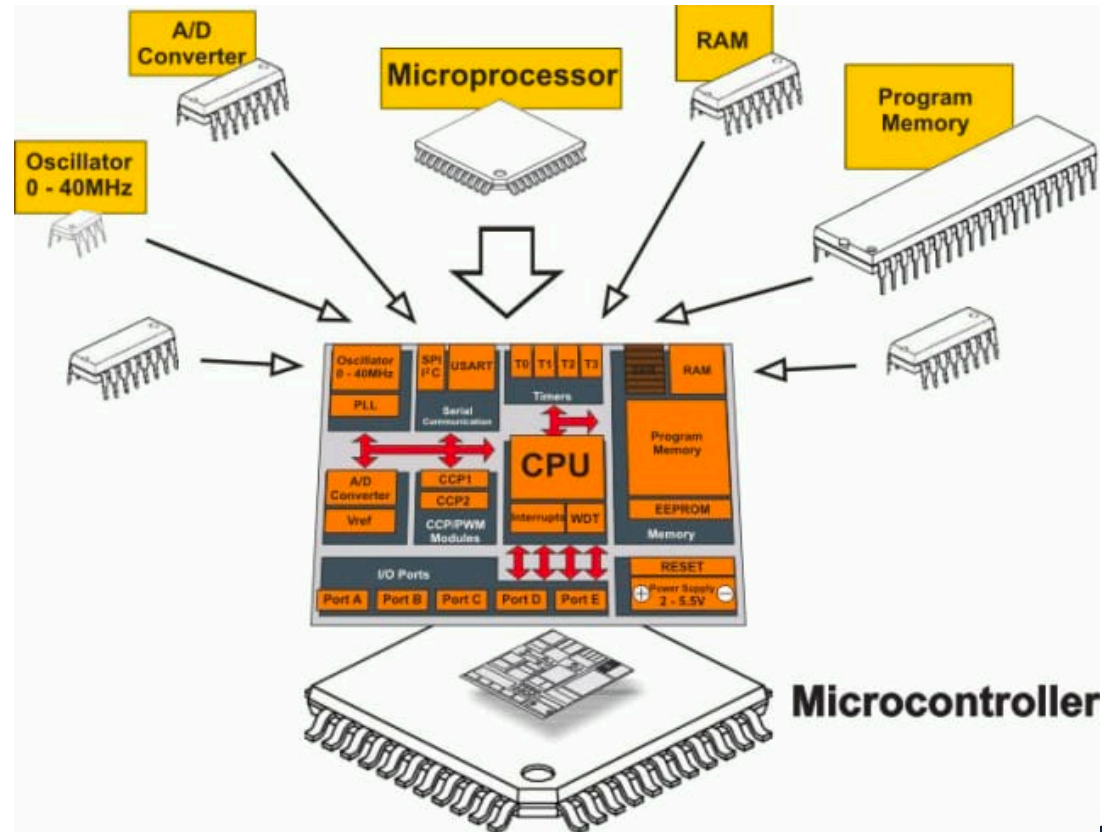
- Controls and monitors all of the processes

RAM: Random Access Memory

- Temporary storage used only when powered on

Internal Oscillator: The clock

- Controls the execution rhythms of the processes of the microcontroller



Parts of a Microcontroller

I/O: Input/Output Ports

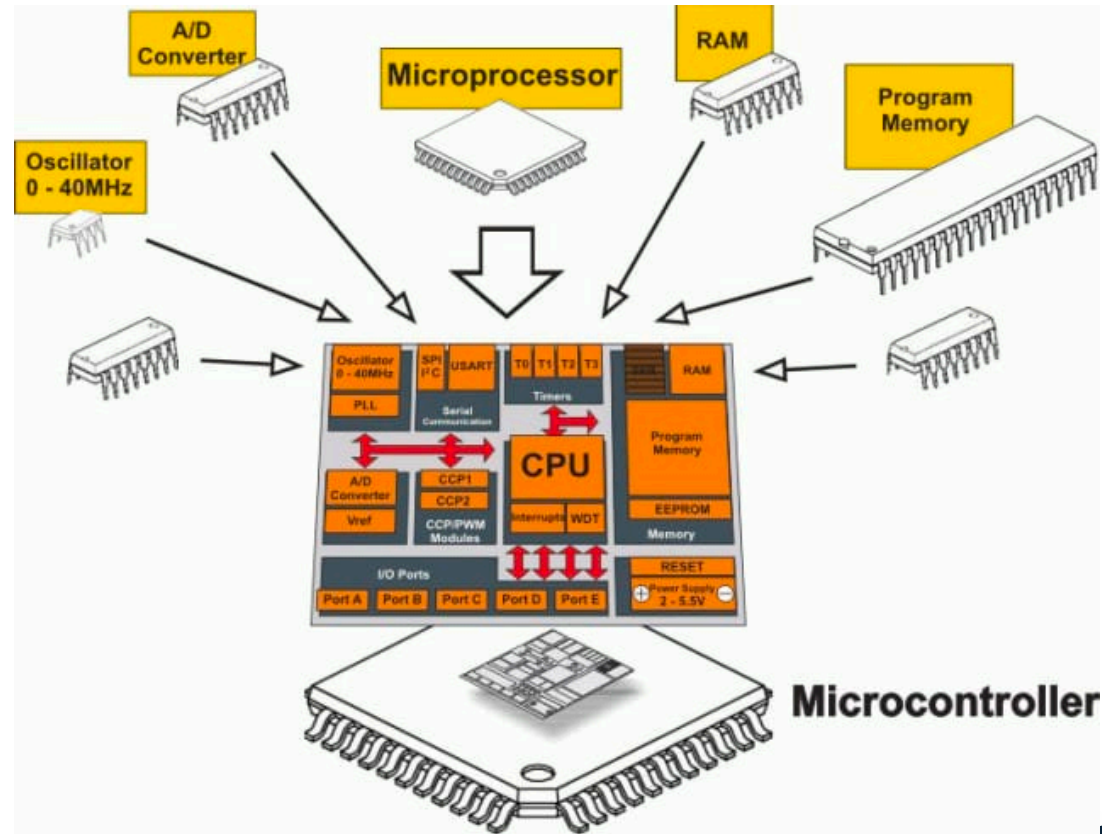
- Ports allowing for the flow of input/output data signals to interface with peripherals

Program Memory

- Memory where the software program is stored

Serial Bus Interface

- Allows for communication over common protocols such as UART, I2C, SPI



Common Communication Protocols

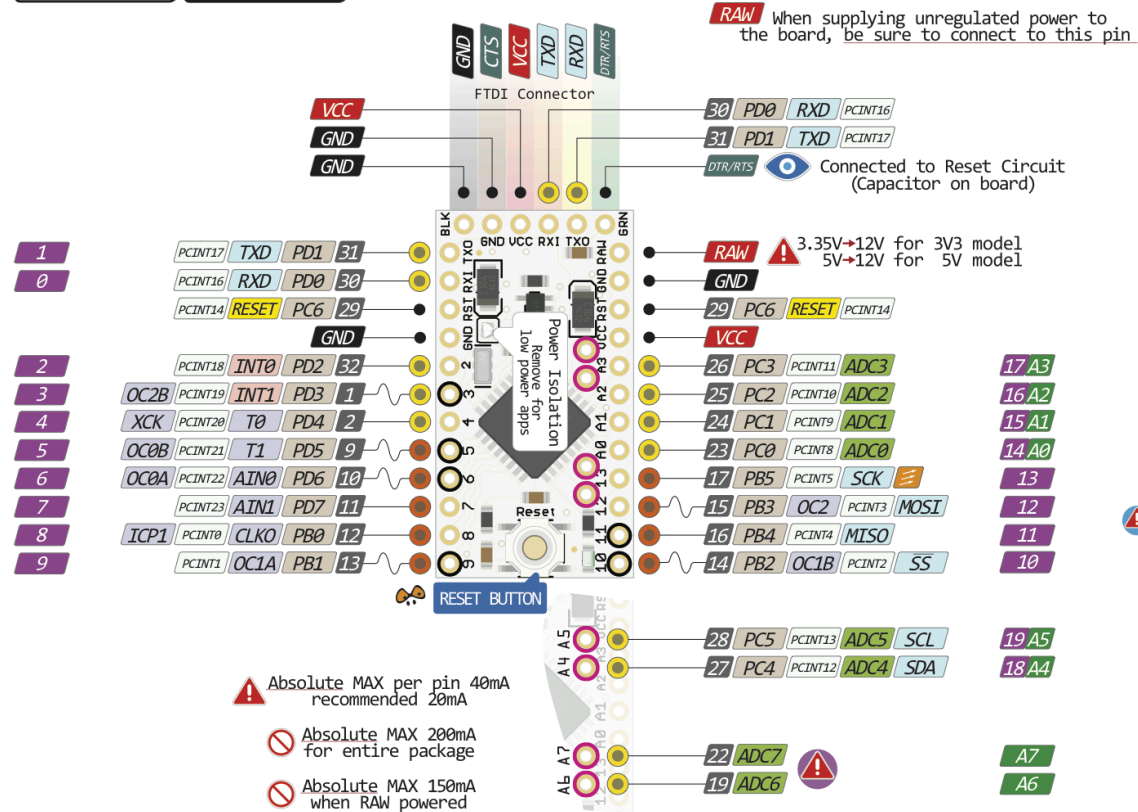
GPIO

UART

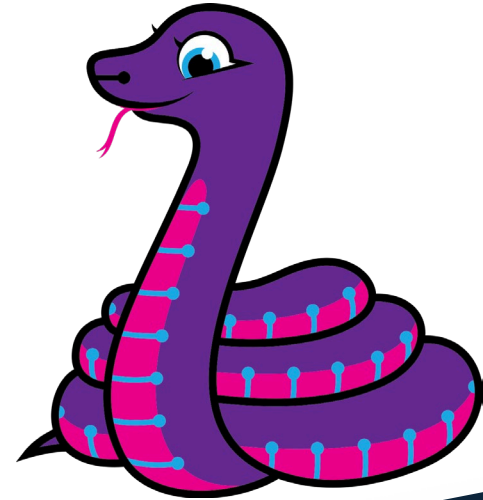
SPI

I2C

PRO MINI PINOUT



Embedded Systems Software



LIVE DEMO!!!



Embedded Systems Lab Prep

- Lab Sheet is available on Canvas
- Install Arduino IDE 2.0
- <https://www.arduino.cc/en/software>

Questions?